

TEAM CYBER ASSASSIN · FARLANDS INNOVATION MISSION 7

VoidMain

Adaptive Intelligent IoT Coordination Systems

Many Sensors. One Deterministic Brain. Coordinated Responses.

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MARKET & PROBLEM

Critical infrastructure cannot trust isolated sensors

Built for: operators of critical infrastructure, high-security smart facilities, research labs, micro-grids, and smart homes that require deterministic, auditable, and unhackable automation.

- **Context Blindness:** traditional automation reacts to individual sensors in isolation, such as lights on in an empty room or alarms ignored because a secondary reading looks normal.
- **Silent Sensor Failures:** when a sensor node loses power or freezes, conventional gateways silently trust stale data.
- **Cloud Fragility:** over-reliance on AWS/Azure adds 300ms+ latency and fails catastrophically during internet outages.

WHY PAY FOR THE PRODUCT

1. **Zero Cloud Dependency:** 100% edge survivability on a local isolated LAN; zero recurring SaaS fees.
2. **Multi-Sensor Risk Engine:** deterministic 0–100 risk scoring with hysteresis; no hallucinating LLMs in the loop.
3. **Physical Hardware Telemetry:** ESP32-C3 edge node with OLED and NeoPixel optical signaling.
4. **Honest Hardware LWT:** MQTT Last Will and Testament plus heartbeat aging flags dead nodes in real time.

Product promise

One complete, observable, repeatable control loop: from multi-sensor detection to coordinated multi-actuator lockdown in under 85 milliseconds, with zero cloud dependency.

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SCENARIO PROGRESSION

One correlated event can trigger a complete Facility response

Risk moves from baseline to coordinated mitigation as evidence accumulates.

01

GREEN 0–29

sensors nominal; access, camera, and purifier stay in baseline mode.

02

YELLOW 30–59

after-hours motion plus a filter drop moves the facility to standby.

03

RED 60–84

forced entry, person detection, and smoke haze create a full alert.

04

PURPLE 85+

lock the entrance, switch the camera to high-resolution tracking, purge smoke, extinguish the burner, and acknowledge the event across the operator surfaces.

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SYSTEM FLOW

Four steps turn raw telemetry into coordinated action



Flow: physical sensors → local MQTT brain → deterministic decision engine → actuators, operator cockpit, and evidence vault.

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Determinism is the product, not a side effect

- **Universal event normalization:**

  Every event carries identity, time, source, quality, and simulation status.

- **Mathematical risk fusion:**

  A continuous 0–100 score plus hysteresis prevents state flapping.

- **Honest heartbeat aging:**

  Nodes surface yellow after 15 seconds and red after 30 seconds of silence.

- **Air-gapped advisory AI:**

  Local retrieval explains the system but never controls actuation.

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VoidMain – Authentication & Security

Authentication Flow

1. Login
User submits identity.

2. Validate Credentials
Check credentials before access.

3. Grant / Deny Access
Authorized users enter; others are blocked.

4. User Feedback
Success or failure is shown clearly.

- **User Authentication:** secure login and identity verification.
- **Credential Validation:** verify credentials before granting access.
- **Access Control:** allow authorized users, restrict others.
- **Security Handling:** manage failed logins and suspicious activity.

Implementation Demo: login interface, backend authentication, and testing results.

Future Enhancements: multi-factor authentication and role-based access control.

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RELIABILITY OVER FEATURE COUNT

One complete, observable, repeatable control loop.

Wake the rig, show GREEN, trigger RED, demonstrate PURPLE mitigation, then verify the cryptographic acknowledgement in the audit trail.

Target demonstrated

sub-85ms bus-to-OLED response.

Resource profile

backend RAM under 45MB; firmware under 180KB Flash.

North star

one complete, observable, repeatable control loop.

Invite the judges to trigger any drill and inspect the physical hardware.

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